PREP 5, LESSONS 9–10

I–V investigations and sensors

Investigating I–V characteristics (required practical); thermistors, LDRs and sensing circuits

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Explain why the resistance of a filament lamp increases with current. DN 102 [1]

A greater current makes the filament hotter, and its resistance increases with temperature.

2 State the function of a diode. DN 77 [1]

To allow current in one direction only.

3 Describe the total resistance of two resistors connected in parallel. DN 636 [1]

It is less than the resistance of the smaller resistor.

Key ideas LESSONS 9–10

Show your working

4 A student finds the I–V characteristic of a lamp.

a Name the independent and dependent variables. [1]

Independent: p.d. across the lamp; dependent: current through it ✓

b State why a variable resistor is included in the circuit. [1]

To change the current (and p.d.) so several readings can be taken ✓

c Explain why the student reverses the connections to the lamp. [1]

To take readings with negative p.d. and current (the other direction) ✓

5 As the temperature of a thermistor increases, its resistance [increases] [decreases] [stays the same] [1]

6 In the dark, the resistance of an LDR is [high] [low] [1]

7 A thermistor is connected in series with a fixed resistor and a battery. The circuit switches on a fan when the room gets hot.

Explain what happens to the current in the thermistor as the room warms up. [2]

The resistance of the thermistor decreases ✓

so the current increases (and can switch on the fan) ✓